

6. The properties of Sea Water

Std 9th

Sub – Geography

Q1. Tick the correct box according to the salinity of the ocean water ✓

(a) Slanting sunrays, melting snow

Ans. Low

(b) Cloudy sky, rainfall throughout the year

Ans. Medium

(c) Clear skies for the most part, perpendicular sunrays

Ans. High

(d) Less supply of freshwater, desert area around

Ans. High

(e) Low temperatures, ample supply of river water

Ans. Low

(f) Continental location, desert around, low rainfall

Ans. High

Q2. Give reasons.

(a) Salinity is low in the land-locked Baltic Sea.

Ans. (i) Baltic Sea is located in temperate zone and is a part of North Atlantic Ocean.

(ii) Baltic Sea is a landlocked sea and hence ocean currents of Atlantic Ocean do not make any change in its properties.

(iii) Despite being a landlocked sea, less temperature, less evaporation and good supply of freshwater due to melting of ice caps resulted in less salinity of Baltic sea.

(b) There is higher salinity in the northern Red Sea while lower in the southern.

Ans. (i) Red sea is an inlet part of an Indian Ocean which is located between the North east Africa and Arabian peninsula.

(ii) It is comparatively a narrow elongated sea.

(iii) The southern end of the sea is towards the ocean while the northern is as good as a landlocked sea.

- (iv) Currents of Indian Ocean largely influence and control the salinity and temperature of southern end.
- (v) Currents do not affect much on northern side of the red sea.
- (vi) Besides, this part is surrounded by Sahara and Arabian deserts.
- (vii) High temperature and more evaporation have resulted in high salinity of the northern part of red sea.

(c) Oceans located at the same latitude do not have same salinity.

Ans. (i) In seas where the rate of evaporation is high than the supply of freshwater salinity is high.

(ii) In seas where the supply of freshwater exceeds the rate of evaporation, salinity is low.

(iii) Salinity is not affected much in seas where both the supply of freshwater and evaporation of water is low.

(iv) As result salinity of sea is not uniform even on the same latitude.

Eg. The salinity of the Indian ocean and latitude ocean is not the same despite of being located on the same latitude

(d) With increasing depth, the temperature of sea water decreases to a certain limit.

Ans. (i) The temperature of the seawater changes rapidly with depth at the equatorial areas. The difference in temperature is lesser in polar areas.

(ii) As a result, the intensity of sunrays decreases with increasing depth. Temperature decreases up to 2000m depth. After 2000m, the temperature of the seawater is uniform everywhere. It is around 4 0C everywhere from the equatorial regions to the polar areas. Temperature reduces only up to 4 0C according to depth. And therefore, the water at greater depths does not freeze.

(iii) While most of the sunrays radiate back from the surface of the sea, some of them penetrate to certain depths in the water.

(e) There are more salt-pans on the Western coast of India than its eastern coast.

Ans. (i) The Arabian sea lies to the west and Bay of Bengal lies to the east of India.

(ii) Many large peninsular river drain their waters in the Bay of Bengal and on the contrary, only small seasonal coastal rivers drain in the Arabian Sea.

(iii) Hence, the salinity of the eastern coast is 34%, while it is 35% in the Arabian Sea.

(iv) Thus, there are more salt pans on the Western coast of India than its Eastern coast.

(f) Salinity increases in the mid-latitudinal zones.

Ans. In temperate regions, the sunrays are slanting and therefore, the temperatures are lower. Because of the melting of the snow, the supply of water is also more, and therefore, in this zone, salinity decreases with increasing latitudes.

Q3. Answer the following questions.

(a) What are the factors affecting the salinity of the sea water?

Ans. (i) In seas where the rate of evaporation is high than the supply of freshwater salinity is high.

(ii) In seas where the supply of freshwater exceeds the rate of evaporation, salinity is low.

(iii) Salinity is not affected much in seas where both the supply of freshwater and evaporation of water is low.

(b) Explain the distribution of salinity around the Tropic of Cancer and tropic of Capricorn.

Ans. (i) The salinity of sea around the Tropic of Cancer is around 36 high temperature.

(ii) High rate of evaporation and landlocked nature of the seas obstructing the free movements of the ocean currents cause more salinity.

(iii) In comparison, the salinity of sea around Tropic of Capricorn is about 35.

(iv) The proportion Of land is less in the southern hemisphere.

(v) As a result Ocean currents control the salinity around Tropic of Capricorn.

(c) What are the factors affecting the temperature of the sea water?

Ans. (i) Temperature is a major property of the seawater. The surface temperature of the seawater is not uniform everywhere. This is dependent on different factors.

(ii) cyclones, rainfall, sea waves, ocean currents, salinity, pollution, convectional currents, seasons, etc. also affect the surface temperature Ocean currents also have similar effects on temperature of the sea water.

(iii) In regions where cold ocean currents flow, the surface temperature of ocean water is less while the regions where the warm currents move, the temperatures increase.

(d) Explain the changes occurring in the temperature of sea water according to the depth.

Ans. While most of the sunrays radiate back from the surface of the sea, some of them penetrate to certain depths in the water. As a result, the intensity of sunrays decreases with increasing depth. Temperature decreases up to 2000m depth. After 2000m, the temperature of the seawater is uniform everywhere. It is around 4°C everywhere from the equatorial regions to the polar areas. Temperature

reduces only up to 4°C according to depth. And therefore, the water at greater depths does not freeze. The temperature of the seawater changes

rapidly with depth at the equatorial areas. The difference in temperature is lesser in polar areas.

(e) Name the factors affecting salinity.

Ans. The factors affecting salinity are:

(i) In seas where the rate of evaporation is high than the supply of freshwater salinity is high.

(ii) In seas where the supply of freshwater exceeds the rate of evaporation, salinity is low.

(iii) Salinity is not affected much in seas where both the supply of freshwater and evaporation of water is low.

Q4. Explain how temperature affects the following.

(a) the density of sea water

Ans. (i) Temperature and salinity are the two properties of sea water that control the density of the sea water. If temperature reduces, density of water increases.

(ii) Cold water is denser and so is saline water. As compared to salinity, temperature affects density more. Hence, sometimes, more saline water has lower temperature at the surface.

(iii) But still, the density of such water is more than other water. On the contrary, seawater having higher temperature and low salinity can have lower density.

(b) the salinity of sea water

Ans. (i) Because of the sun's heat, evaporation happens at a faster rate.

(ii) Evaporated water turns into water vapour and reduces in quantity. But amount of salt remains the same in the remaining water and therefore the salinity of the water increases.

(iii) In seas where the rate of evaporation is high than the supply of freshwater salinity is high.

(iv) In seas where the supply of freshwater exceeds the rate of evaporation, salinity is low.

(v) Salinity is not affected much in seas where both the supply of freshwater and evaporation of water is low.